

Aditya Karan

+91 9942116031 | adityakr.sinha2661@gmail.com | ML Engineer | Computer Vision | Geospatial AI | IoT Systems

SUMMARY

Machine Learning & Data Science Engineer with hands-on experience in computer vision, geospatial AI, and IoT-integrated detection systems. Developed and deployed end-to-end YOLOv8-based drone detection pipelines and geospatial anomaly models for defense-grade applications at DRDO. Published researcher (IJERT/Zenodo). Pursuing B.Tech in Computer Engineering (CGPA: 8.88). Passionate about applying ML research to real-world, high-impact problems.

TECHNICAL SKILLS

- **Languages & Databases:** Python, SQL **Libraries:** NumPy, Pandas, Matplotlib, Seaborn
- **ML & Deep Learning:** YOLOv8, Scikit-learn, Model Training & Evaluation, Data Preprocessing, Feature Engineering
- **Computer Vision:** OpenCV, Object Detection, Motion Analysis, Real-Time Video Processing
- **Geospatial & Remote Sensing:** GIS (ArcGIS, QGIS), Google Earth Engine, Satellite Image Analysis, Land-Change Detection
- **Tools & Platforms:** Jupyter Notebook, Git, Flask, ESP32 (IoT), Telegram API **Competencies:** Research Documentation, End-to-End ML Pipelines

EXPERIENCE

Research Fellow – Geospatial AI & Remote Sensing

Jun 2024 – Aug 2024

Defence Research and Development Lab (DRDL), DRDO India

- Built geospatial anomaly detection models for border surveillance using satellite imagery and multi-temporal land-change analysis with ArcGIS, QGIS, and Google Earth Engine.
- Performed feature extraction, risk modelling, and data preprocessing on large-scale remote sensing datasets to support defense intelligence workflows.
- Co-authored research published in IJERT (DOI: 10.5281/zenodo.18441874); delivered technical reports for the DRDO project team.

DRDO Representative – Industrial Training

Jun 2024

Low Emissions Carbon Pathway Program India

- Represented DRDO in cross-organizational technical sessions on sustainable development, environmental impact assessment, and low-emission strategies.
- Contributed to collaborative discussions on carbon reduction pathways with multi-disciplinary engineering teams.

PROJECTS

Scenario-Driven Land Change Anomaly Assessment for Border Surveillance

DRDO, Jul 2024 – Aug 2025 / Published

- Designed an ML pipeline for detecting land-cover anomalies in satellite imagery using GIS and Remote Sensing tools for defense border surveillance.
- Applied multi-temporal feature extraction and spatial risk modelling across large geospatial datasets to identify statistically significant change patterns.
- Research published in IJERT; full citation at DOI: zenodo.18441874.

Physics-Informed IoT-Based Real-Time Drone Detection System

Python, YOLOv8, OpenCV, ESP32

- Engineered a real-time drone detection system using YOLOv8 and OpenCV on live video streams, achieving high precision in distinguishing drones from birds via physics-informed motion filtering (speed, trajectory, stability).
- Reduced false positive rate significantly by integrating kinematic constraints into the object classification post-processing stage.
- Deployed an IoT alert layer using ESP32 microcontrollers and Telegram API for automated real-time threat notifications.
- Built a complete end-to-end pipeline covering video ingestion, detection, classification, and alert dispatch; validated on real-world footage and a custom-annotated dataset.

EDUCATION

Aditya Silver Oak Institute of Technology

2022 – 2026

B.Tech in Computer Engineering | **CGPA: 8.88 / 10**

Ahmedabad, India

The Earth Public School

2019 – 2022

Higher Secondary Certificate (HSC), Science Stream

PUBLICATIONS & ACHIEVEMENTS

- **Published:** “Scenario-Driven Land Changes Anomaly Assessment for Border Surveillance,” IJERT, 2025. DOI: 10.5281/zenodo.18441874
- **Defense Collaboration:** Selected as Research Fellow at DRDL (DRDO) while in 3rd year of undergraduate studies.
- **Academic Standing:** Maintained CGPA of 8.88/10 across the Computer Engineering program.